

# Package ‘fitVARMxID’

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**Title** Fit the Vector Autoregressive Model for Multiple Individuals

**Version** 1.0.2.9000

**Description** Fit the vector autoregressive model for multiple individuals using the 'OpenMx' package (Hunter, 2017 <[doi:10.1080/10705511.2017.1369354](https://doi.org/10.1080/10705511.2017.1369354)>).

**URL** <https://github.com/jeksterslab/fitVARMxID>,  
<https://jeksterslab.github.io/fitVARMxID/>

**BugReports** <https://github.com/jeksterslab/fitVARMxID/issues>

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**Encoding** UTF-8

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**NeedsCompilation** no

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coef.varmxid

*Parameter Estimates***Description**

Parameter Estimates

**Usage**

```
## S3 method for class 'varmxid'
coef(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  ncores = NULL,
  ...
)
```

**Arguments**

|        |                                                                                                                                         |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class varmxid.                                                                                                                |
| mu     | Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.             |
| alpha  | Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector. |
| beta   | Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.     |
| nu     | Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.             |
| psi    | Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.         |
| theta  | Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix. |
| ncores | Positive integer. Number of cores to use.                                                                                               |
| ...    | additional arguments.                                                                                                                   |

**Value**

Returns a list of vectors of parameter estimates.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

converged

*Check Model Convergence*

---

**Description**

Determines whether each fitted **OpenMx** model in a `varmxid` object meets convergence criteria based on (a) acceptable optimizer status and gradient size, (b) a positive-definite Hessian, and (c) parameters not being at their bounds.

**Usage**

```
converged(object, ...)
```

```
## S3 method for class 'varmxid'  
converged(object, prop = FALSE, ...)
```

**Arguments**

|                     |                                                                                                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code> | A fit object.                                                                                                                                                                                                       |
| <code>...</code>    | Passed to and/or used by methods.                                                                                                                                                                                   |
| <code>prop</code>   | Logical. If <code>prop = FALSE</code> , a named logical vector indicating, for each individual fit, whether the convergence criteria are met. If <code>prop = TRUE</code> , the proportion of cases that converged. |

**Value**

For the `varmxid` method: If `prop = FALSE`, a named logical vector indicating, for each individual fit, whether the convergence criteria are met. If `prop = TRUE`, the proportion of cases that converged.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

FitVARMxID

*Fit the First-Order Vector Autoregressive Model by ID*


---

### Description

The function fits the first-order vector autoregressive model for each unit ID.

### Usage

```
FitVARMxID(
  data,
  observed,
  id,
  time = NULL,
  ct = FALSE,
  center = TRUE,
  mu_fixed = FALSE,
  mu_free = NULL,
  mu_values = NULL,
  mu_lbound = NULL,
  mu_ubound = NULL,
  alpha_fixed = FALSE,
  alpha_free = NULL,
  alpha_values = NULL,
  alpha_lbound = NULL,
  alpha_ubound = NULL,
  beta_fixed = FALSE,
  beta_free = NULL,
  beta_values = NULL,
  beta_lbound = NULL,
  beta_ubound = NULL,
  psi_diag = FALSE,
  psi_fixed = FALSE,
  psi_d_free = NULL,
  psi_d_values = NULL,
  psi_d_lbound = NULL,
  psi_d_ubound = NULL,
  psi_d_equal = FALSE,
  psi_l_free = NULL,
  psi_l_values = NULL,
  psi_l_lbound = NULL,
  psi_l_ubound = NULL,
  nu_fixed = TRUE,
  nu_free = NULL,
  nu_values = NULL,
  nu_lbound = NULL,
  nu_ubound = NULL,
```

```

theta_diag = TRUE,
theta_fixed = TRUE,
theta_d_free = NULL,
theta_d_values = NULL,
theta_d_lbound = NULL,
theta_d_ubound = NULL,
theta_d_equal = FALSE,
theta_l_free = NULL,
theta_l_values = NULL,
theta_l_lbound = NULL,
theta_l_ubound = NULL,
mu0_fixed = TRUE,
mu0_func = TRUE,
mu0_free = NULL,
mu0_values = NULL,
mu0_lbound = NULL,
mu0_ubound = NULL,
sigma0_fixed = TRUE,
sigma0_func = TRUE,
sigma0_diag = FALSE,
sigma0_d_free = NULL,
sigma0_d_values = NULL,
sigma0_d_lbound = NULL,
sigma0_d_ubound = NULL,
sigma0_d_equal = FALSE,
sigma0_l_free = NULL,
sigma0_l_values = NULL,
sigma0_l_lbound = NULL,
sigma0_l_ubound = NULL,
robust = FALSE,
seed = NULL,
tries_explore = 100,
tries_local = 100,
max_attempts = 10,
silent = FALSE,
ncores = NULL
)

```

### Arguments

|          |                                                                                                                                                                                           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data     | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), and at least one column of observed values. |
| observed | Character vector. A vector of character strings of the names of the observed variables in the data.                                                                                       |
| id       | Character string. A character string of the name of the ID variable in the data.                                                                                                          |
| time     | Character string. A character string of the name of the TIME variable in the data. Used when ct = TRUE.                                                                                   |

|              |                                                                                                                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ct           | Logical. If TRUE, fit a continuous-time vector autoregressive model. If FALSE, fit a discrete-time vector autoregressive model.                                                                                               |
| center       | Logical. If TRUE, use the mean-centered (mean-reverting) state equation. When center = TRUE, alpha is implied and the set-point mu is estimated. When center = FALSE, alpha is estimated and mu is implied.                   |
| mu_fixed     | Logical. If TRUE, the set-point mean vector mu is fixed to mu_values. If mu_fixed = TRUE and mu_values = NULL, mu is fixed to a zero vector. If FALSE, mu is estimated.                                                       |
| mu_free      | Logical vector indicating which elements of mu are freely estimated. If NULL, all elements are free. Ignored if mu_fixed = TRUE.                                                                                              |
| mu_values    | Numeric vector of values for mu. If mu_fixed = TRUE, these are fixed values. If mu_fixed = FALSE, these are starting values. If NULL, defaults to a vector of zeros.                                                          |
| mu_lbound    | Numeric vector of lower bounds for mu. Ignored if mu_fixed = TRUE.                                                                                                                                                            |
| mu_ubound    | Numeric vector of upper bounds for mu. Ignored if mu_fixed = TRUE.                                                                                                                                                            |
| alpha_fixed  | Logical. If TRUE, the dynamic model intercept vector alpha is fixed to alpha_values. If FALSE, alpha is estimated.                                                                                                            |
| alpha_free   | Logical vector indicating which elements of alpha are freely estimated. If NULL, all elements are free. Ignored if alpha_fixed = TRUE.                                                                                        |
| alpha_values | Numeric vector of values for alpha. If alpha_fixed = TRUE, these are fixed values. If alpha_fixed = FALSE, these are starting values. If NULL, defaults to a vector of zeros.                                                 |
| alpha_lbound | Numeric vector of lower bounds for alpha. Ignored if alpha_fixed = TRUE.                                                                                                                                                      |
| alpha_ubound | Numeric vector of upper bounds for alpha. Ignored if alpha_fixed = TRUE.                                                                                                                                                      |
| beta_fixed   | Logical. If TRUE, the dynamic model coefficient matrix beta is fixed. If FALSE, beta is estimated.                                                                                                                            |
| beta_free    | Logical matrix indicating which elements of beta are freely estimated. If NULL, all elements are free. Ignored if beta_fixed = TRUE.                                                                                          |
| beta_values  | Numeric matrix. Values for beta. If beta_fixed = TRUE, these are fixed values; if beta_fixed = FALSE, these are starting values. If NULL, defaults to a diagonal matrix with -0.001 when ct = TRUE and 0.001 when ct = FALSE. |
| beta_lbound  | Numeric matrix of lower bounds for beta. If NULL, defaults to -2.5. Ignored if beta_fixed = TRUE.                                                                                                                             |
| beta_ubound  | Numeric matrix. Upper bounds for beta. Ignored if beta_fixed = TRUE. If NULL, defaults to +2.5. If NULL and ct = TRUE, diagonal upper bounds are set to -1e-05.                                                               |
| psi_diag     | Logical. If TRUE, psi is diagonal. If FALSE, psi is symmetric.                                                                                                                                                                |
| psi_fixed    | Logical. If TRUE, the process noise covariance matrix psi is fixed using psi_d_values and psi_l_values. If psi_d_values is NULL it is fixed to a zero matrix. If FALSE, psi is estimated.                                     |
| psi_d_free   | Logical vector indicating free/fixed status of the elements of psi_d. If NULL, all element of psi_d are free.                                                                                                                 |

|                |                                                                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| psi_d_values   | Numeric vector with starting values for psi_d. If psi_fixed = TRUE, these are fixed values. If psi_fixed = FALSE, these are starting values.                                                            |
| psi_d_lbound   | Numeric vector with lower bounds for psi_d.                                                                                                                                                             |
| psi_d_ubound   | Numeric vector with upper bounds for psi_d.                                                                                                                                                             |
| psi_d_equal    | Logical. When TRUE, all free diagonal elements of psi_d are constrained to be equal and estimated as a single shared parameter. Ignored if no diagonal elements are free.                               |
| psi_l_free     | Logical matrix indicating which strictly-lower-triangular elements of psi_l are free. If NULL, all element of psi_l are free. Ignored if psi_diag = TRUE.                                               |
| psi_l_values   | Numeric matrix of starting values for the strictly-lower-triangular elements of psi_l.                                                                                                                  |
| psi_l_lbound   | Numeric matrix with lower bounds for psi_l.                                                                                                                                                             |
| psi_l_ubound   | Numeric matrix with upper bounds for psi_l.                                                                                                                                                             |
| nu_fixed       | Logical. If TRUE, the measurement model intercept vector nu is fixed to nu_values. If FALSE, nu is estimated.                                                                                           |
| nu_free        | Logical vector indicating which elements of nu are freely estimated. If NULL, all elements are free. Ignored if nu_fixed = TRUE.                                                                        |
| nu_values      | Numeric vector of values for nu. If nu_fixed = TRUE, these are fixed values. If nu_fixed = FALSE, these are starting values. If NULL, defaults to a vector of zeros.                                    |
| nu_lbound      | Numeric vector of lower bounds for nu. Ignored if nu_fixed = TRUE.                                                                                                                                      |
| nu_ubound      | Numeric vector of upper bounds for nu. Ignored if nu_fixed = TRUE.                                                                                                                                      |
| theta_diag     | Logical. If TRUE, theta is diagonal. If FALSE, theta is symmetric.                                                                                                                                      |
| theta_fixed    | Logical. If TRUE, the measurement error covariance matrix theta is fixed using theta_d_values and theta_l_values. If theta_d_values is NULL it is fixed to a zero matrix. If FALSE, theta is estimated. |
| theta_d_free   | Logical vector indicating free/fixed status of the diagonal parameters theta_d. If NULL, all element of theta_d are free.                                                                               |
| theta_d_values | Numeric vector with starting values for theta_d. If theta_fixed = TRUE, these are fixed values. If theta_fixed = FALSE, these are starting values.                                                      |
| theta_d_lbound | Numeric vector with lower bounds for theta_d.                                                                                                                                                           |
| theta_d_ubound | Numeric vector with upper bounds for theta_d.                                                                                                                                                           |
| theta_d_equal  | Logical. When TRUE, all free diagonal elements of theta_d are constrained to be equal and estimated as a single shared parameter. Ignored if no diagonal elements are free.                             |
| theta_l_free   | Logical matrix indicating which strictly-lower-triangular elements of theta_l are free. If NULL, all element of theta_l are free. Ignored if theta_diag = TRUE.                                         |
| theta_l_values | Numeric matrix of starting values for the strictly-lower-triangular elements of theta_l.                                                                                                                |
| theta_l_lbound | Numeric matrix with lower bounds for theta_l.                                                                                                                                                           |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>theta_l_ubound</code>  | Numeric matrix with upper bounds for <code>theta_l</code> .                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>mu0_fixed</code>       | Logical. If TRUE, the initial mean vector <code>mu0</code> is fixed. If <code>mu0_fixed</code> = TRUE and <code>mu0_func</code> = TRUE, <code>mu0</code> is fixed to the implied stable mean vector. If <code>mu0_fixed</code> = TRUE and <code>mu0_values</code> = NULL, <code>mu0</code> is fixed to a zero vector. If FALSE, <code>mu0</code> is estimated.                                                                              |
| <code>mu0_func</code>        | Logical. If TRUE and <code>mu0_fixed</code> = TRUE, <code>mu0</code> is fixed to the implied stable mean vector.                                                                                                                                                                                                                                                                                                                            |
| <code>mu0_free</code>        | Logical vector indicating which elements of <code>mu0</code> are free. Ignored if <code>mu0_fixed</code> = TRUE.                                                                                                                                                                                                                                                                                                                            |
| <code>mu0_values</code>      | Numeric vector of values for <code>mu0</code> . If <code>mu0_fixed</code> = TRUE, these are fixed values. If <code>mu0_fixed</code> = FALSE, these are starting values. If NULL, defaults to a vector of zeros. Ignored if <code>mu0_fixed</code> = TRUE and <code>mu0_func</code> = TRUE.                                                                                                                                                  |
| <code>mu0_lbound</code>      | Numeric vector of lower bounds for <code>mu0</code> . Ignored if <code>mu0_fixed</code> = TRUE.                                                                                                                                                                                                                                                                                                                                             |
| <code>mu0_ubound</code>      | Numeric vector of upper bounds for <code>mu0</code> . Ignored if <code>mu0_fixed</code> = TRUE.                                                                                                                                                                                                                                                                                                                                             |
| <code>sigma0_fixed</code>    | Logical. If TRUE, the initial condition covariance matrix <code>sigma0</code> is fixed using <code>sigma0_d_values</code> and <code>sigma0_l_values</code> . If <code>sigma0_fixed</code> = TRUE and <code>sigma0_func</code> = TRUE, <code>sigma0</code> is fixed to the implied stable covariance matrix. If <code>sigma0_fixed</code> = TRUE and <code>sigma0_d_values</code> = NULL, <code>sigma0</code> is fixed to a diffused matrix. |
| <code>sigma0_func</code>     | Logical. If TRUE and <code>sigma0_fixed</code> = TRUE, <code>sigma0</code> is fixed to the implied stable covariance matrix.                                                                                                                                                                                                                                                                                                                |
| <code>sigma0_diag</code>     | Logical. If TRUE, <code>sigma0</code> is diagonal. If FALSE, <code>sigma0</code> is symmetric.                                                                                                                                                                                                                                                                                                                                              |
| <code>sigma0_d_free</code>   | Logical vector indicating free/fixed status of the elements of <code>sigma0_d</code> . If NULL, all element of <code>sigma0_d</code> are free.                                                                                                                                                                                                                                                                                              |
| <code>sigma0_d_values</code> | Numeric vector with starting values for <code>sigma0_d</code> . If <code>sigma0_fixed</code> = TRUE, these are fixed values. If <code>sigma0_fixed</code> = FALSE, these are starting values.                                                                                                                                                                                                                                               |
| <code>sigma0_d_lbound</code> | Numeric vector with lower bounds for <code>sigma0_d</code> .                                                                                                                                                                                                                                                                                                                                                                                |
| <code>sigma0_d_ubound</code> | Numeric vector with upper bounds for <code>sigma0_d</code> .                                                                                                                                                                                                                                                                                                                                                                                |
| <code>sigma0_d_equal</code>  | Logical. When TRUE, all free diagonal elements of <code>sigma0_d</code> are constrained to be equal and estimated as a single shared parameter. Ignored if no diagonal elements are free.                                                                                                                                                                                                                                                   |
| <code>sigma0_l_free</code>   | Logical matrix indicating which strictly-lower-triangular elements of <code>sigma0_l</code> are free. If NULL, all element of <code>sigma0_l</code> are free. Ignored if <code>sigma0_diag</code> = TRUE.                                                                                                                                                                                                                                   |
| <code>sigma0_l_values</code> | Numeric matrix of starting values for the strictly-lower-triangular elements of <code>sigma0_l</code> .                                                                                                                                                                                                                                                                                                                                     |
| <code>sigma0_l_lbound</code> | Numeric matrix with lower bounds for <code>sigma0_l</code> .                                                                                                                                                                                                                                                                                                                                                                                |
| <code>sigma0_l_ubound</code> | Numeric matrix with upper bounds for <code>sigma0_l</code> .                                                                                                                                                                                                                                                                                                                                                                                |



|               |                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------|
| robust        | Logical. If TRUE, calculate robust (sandwich) sampling variance-covariance matrix.                      |
| seed          | Random seed for reproducibility.                                                                        |
| tries_explore | Integer. Number of extra tries for the wide exploration phase.                                          |
| tries_local   | Integer. Number of extra tries for local polishing.                                                     |
| max_attempts  | Integer. Maximum number of remediation attempts after the first Hessian computation fails the criteria. |
| silent        | Logical. If TRUE, suppresses messages during the model fitting stage.                                   |
| ncores        | Positive integer. Number of cores to use.                                                               |

## Details

### Measurement Model:

By default, the measurement model is given by

$$\mathbf{y}_{i,t} = \boldsymbol{\eta}_{i,t}.$$

However, the full measurement model can be parameterized as follows

$$\mathbf{y}_{i,t} = \boldsymbol{\nu}_i + \boldsymbol{\Lambda} \boldsymbol{\eta}_{i,t} + \boldsymbol{\varepsilon}_{i,t}, \quad \text{with} \quad \boldsymbol{\varepsilon}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}_i)$$

where  $\mathbf{y}_{i,t}$ ,  $\boldsymbol{\eta}_{i,t}$ , and  $\boldsymbol{\varepsilon}_{i,t}$  are random variables and  $\boldsymbol{\nu}_i$ ,  $\boldsymbol{\Lambda}$ , and  $\boldsymbol{\Theta}_i$  are model parameters.  $\mathbf{y}_{i,t}$  represents a vector of observed random variables,  $\boldsymbol{\eta}_{i,t}$  a vector of latent random variables, and  $\boldsymbol{\varepsilon}_{i,t}$  a vector of random measurement errors, at time  $t$  and individual  $i$ .  $\boldsymbol{\nu}_i$  denotes a vector of intercepts (fixed to a null vector by default),  $\boldsymbol{\Lambda}$  a matrix of factor loadings, and  $\boldsymbol{\Theta}_i$  the covariance matrix of  $\boldsymbol{\varepsilon}$ . In this model,  $\boldsymbol{\Lambda}$  is an identity matrix and  $\boldsymbol{\Theta}_i$  is a diagonal matrix.

### Discrete-Time Dynamic Structure:

The dynamic structure is given by

$$\boldsymbol{\eta}_{i,t} = \boldsymbol{\alpha}_i + \boldsymbol{\beta}_i \boldsymbol{\eta}_{i,t-1} + \boldsymbol{\zeta}_{i,t}, \quad \text{with} \quad \boldsymbol{\zeta}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi}_i)$$

where  $\boldsymbol{\eta}_{i,t}$ ,  $\boldsymbol{\eta}_{i,t-1}$ , and  $\boldsymbol{\zeta}_{i,t}$  are random variables, and  $\boldsymbol{\alpha}_i$ ,  $\boldsymbol{\beta}_i$ , and  $\boldsymbol{\Psi}_i$  are model parameters. Here,  $\boldsymbol{\eta}_{i,t}$  is a vector of latent variables at time  $t$  and individual  $i$ ,  $\boldsymbol{\eta}_{i,t-1}$  represents a vector of latent variables at time  $t-1$  and individual  $i$ , and  $\boldsymbol{\zeta}_{i,t}$  represents a vector of dynamic noise at time  $t$  and individual  $i$ .  $\boldsymbol{\alpha}_i$  denotes a vector of intercepts,  $\boldsymbol{\beta}_i$  a matrix of autoregression and cross regression coefficients, and  $\boldsymbol{\Psi}_i$  the covariance matrix of  $\boldsymbol{\zeta}_{i,t}$ .

If center = TRUE, the dynamic structure is parameterized as follows

$$\boldsymbol{\eta}_{i,t} = \boldsymbol{\mu}_i + \boldsymbol{\beta}_i (\boldsymbol{\eta}_{i,t-1} - \boldsymbol{\mu}_i) + \boldsymbol{\zeta}_{i,t}$$

where  $\boldsymbol{\mu}_i$  is equilibrium level of the latent state toward which the system is pulled over time.

### Continuous-Time Dynamic Structure:

The continuous-time parameterization, when ct = TRUE, for the dynamic structure is given by

$$d\boldsymbol{\eta}_{i,t} = (\boldsymbol{\alpha}_i + \boldsymbol{\beta}_i \boldsymbol{\eta}_{i,t}) dt + \boldsymbol{\Psi}_i^{\frac{1}{2}} d\mathbf{W}_{i,t}$$

note that  $d\mathbf{W}$  is a Wiener process or Brownian motion, which represents random fluctuations.

If center = TRUE, the dynamic structure is parameterized as follows

$$d\boldsymbol{\eta}_{i,t} = \boldsymbol{\beta}_i (\boldsymbol{\eta}_{i,t} - \boldsymbol{\mu}_i) dt + \boldsymbol{\Psi}_i^{\frac{1}{2}} d\mathbf{W}_{i,t}$$

**Value**

Returns an object of class `varmxid` which is a list with the following elements:

**call** Function call.

**args** List of function arguments.

**fun** Function used ("FitVARMxID").

**model** A list of generated OpenMx models.

**output** A list of fitted OpenMx models.

**converged** A logical vector indicating converged cases.

**robust** A list of output from `OpenMx::mxRobustSE()` with argument `details = TRUE` for each `id` if `robust = TRUE`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Hunter, M. D. (2017). State space modeling in an open source, modular, structural equation modeling environment. *Structural Equation Modeling: A Multidisciplinary Journal*, 25(2), 307–324. doi:10.1080/10705511.2017.1369354

Neale, M. C., Hunter, M. D., Pritikin, J. N., Zahery, M., Brick, T. R., Kirkpatrick, R. M., Estabrook, R., Bates, T. C., Maes, H. H., & Boker, S. M. (2015). OpenMx 2.0: Extended structural equation and statistical modeling. *Psychometrika*, 81(2), 535–549. doi:10.1007/s1133601494358

**See Also**

Other VAR Functions: `LDL()`, `Softplus()`

**Examples**

```
# Generate data using the simStateSpace package-----
library(simStateSpace)
set.seed(42)
n <- 5
time <- 100
p <- 2
alpha <- rep(x = 0, times = p)
beta <- 0.50 * diag(p)
psi <- 0.001 * diag(p)
psi_l <- t(chol(psi))
mu0 <- SSMMeanEta(
  beta = beta,
  alpha = alpha
)
sigma0 <- SSMCovEta(
  beta = beta,
  psi = psi
```

```

)
sigma0_l <- t(chol(sigma0))
sim <- SimSSMVARFixed(
  n = n,
  time = time,
  mu0 = mu0,
  sigma0_l = sigma0_l,
  alpha = alpha,
  beta = beta,
  psi_l = psi_l
)
data <- as.data.frame(sim)

# Fit the model-----
# center = TRUE
library(fitVARMxID)
fit <- FitVARMxID(
  data = data,
  observed = paste0("y", seq_len(p)),
  id = "id",
  center = TRUE
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)
converged(fit)

# Fit the model-----
# center = FALSE
library(fitVARMxID)
fit <- FitVARMxID(
  data = data,
  observed = paste0("y", seq_len(p)),
  id = "id",
  center = FALSE
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)
converged(fit)

```

**Description**

Performs an LDL' factorization of a symmetric positive-definite matrix  $X$ , such that

$$X = LDL',$$

where  $L$  is unit lower-triangular (ones on the diagonal) and  $D$  is diagonal.

**Usage**

```
LDL(x, epsilon = 1e-10)
```

```
InvLDL(s_l, uc_d)
```

**Arguments**

|                      |                                                                                                                                                                         |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>       | Numeric matrix. Assumed symmetric positive-definite (not checked). Note: <code>LDL()</code> may error if the implied diagonal entries of $D$ are not strictly positive. |
| <code>epsilon</code> | Numeric. Small positive value used to replace <i>exactly zero</i> diagonal entries of $x$ prior to factorization.                                                       |
| <code>s_l</code>     | Matrix. Strictly lower-triangular part of $L$ . In <code>InvLDL()</code> , only the strictly lower triangle is used (upper triangle and diagonal are ignored).          |
| <code>uc_d</code>    | Vector. Unconstrained vector such that <code>Softplus(uc_d) = d</code> , where $d$ are the diagonal entries of $D$ .                                                    |

**Details**

`LDL()` returns both the unit lower-triangular factor  $L$  and the diagonal factor  $D$ . The strictly lower-triangular part of  $L$  is also provided for convenience. The function additionally computes an unconstrained vector `uc_d` such that `Softplus(uc_d) = d`. This uses a numerically stable inverse softplus implementation based on `log(expm1(d))` (and a large- $d$  rewrite), rather than the unstable expression `log(exp(d) - 1)`.

`InvLDL()` returns a symmetric positive definite matrix from the strictly lower-triangular part of  $L$  and the unconstrained vector `uc_d`. The reconstructed matrix is symmetrized as  $(\Sigma + \Sigma')/2$  to reduce numerical asymmetry.

**Value**

- `LDL()`: a list with components:
  - `l`: a unit lower-triangular matrix  $L$
  - `s_l`: a strictly lower-triangular part of  $L$
  - `d`: a vector of diagonal entries of  $D$
  - `uc_d`: unconstrained vector with `softplus(uc_d) = d`
  - `x`: input matrix (with diagonal zeros possibly replaced by `epsilon`)
  - `epsilon`: the `epsilon` value used
- `InvLDL()`: a symmetric positive definite matrix

**See Also**

Other VAR Functions: [FitVARmXID\(\)](#), [Softplus\(\)](#)

**Examples**

```
set.seed(123)
x <- crossprod(matrix(rnorm(16), 4, 4)) + diag(1e-6, 4)
ldl <- LDL(x = x)
ldl
inv_ldl <- InvLDL(s_l = ldl$s_l, uc_d = ldl$uc_d)
inv_ldl
max(abs(x - inv_ldl))
```

---

|               |                                                 |
|---------------|-------------------------------------------------|
| print.varmxid | <i>Print Method for Object of Class varmxid</i> |
|---------------|-------------------------------------------------|

---

**Description**

Print Method for Object of Class varmxid

**Usage**

```
## S3 method for class 'varmxid'
print(
  x,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ...
)
```

**Arguments**

|       |                                                                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------|
| x     | an object of class varmxid.                                                                                                             |
| means | Logical. If means = TRUE, return means. Otherwise, the function returns raw estimates.                                                  |
| mu    | Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.             |
| alpha | Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector. |

|        |                                                                                                                                         |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------|
| beta   | Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.     |
| nu     | Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.             |
| psi    | Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.         |
| theta  | Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix. |
| digits | Integer indicating the number of decimal places to display.                                                                             |
| ...    | further arguments.                                                                                                                      |

**Value**

Prints means or raw estimates depending the the value of the argument means.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

Softplus

*Softplus and Inverse Softplus Transformations*


---

**Description**

The softplus transformation maps unconstrained real values to the positive real line. This is useful when parameters (e.g., variances) must be positive. The inverse softplus transformation recovers the unconstrained value from a strictly positive input.

**Usage**

Softplus(x)

InvSoftplus(x)

**Arguments**

x                      Numeric vector or matrix. Input values to be transformed.

**Details**

Mathematical definitions:

- $\text{Softplus}(x) = \log(1 + \exp(x))$
- $\text{InvSoftplus}(x) = \log(\exp(x) - 1)$

Numerical implementation (stable forms):

- Softplus( $x$ ) =  $\max(x, 0) + \log1p(\exp(-\text{abs}(x)))$
- InvSoftplus( $y$ ) uses  $\log(\expm1(y))$ , and for large  $y$  uses the rewrite  $y + \log1p(-\exp(-y))$  for stability.

For numerical stability, these functions use `log1p()` and `expm1()` internally. `InvSoftplus()` requires strictly positive input and will error if any values are  $\leq 0$ .

### Value

- `Softplus()`: numeric vector or matrix of nonnegative values (mathematically strictly positive for finite inputs, but can underflow to 0 for very negative values).
- `InvSoftplus()`: numeric vector or matrix of unconstrained values.

### Author(s)

Ivan Jacob Agaloos Pesigan

### See Also

Other VAR Functions: [FitVARMxID\(\)](#), [LDL\(\)](#)

### Examples

```
# Apply softplus to unconstrained values
x <- c(-5, 0, 5)
y <- Softplus(x)

# Recover unconstrained values
x_recovered <- InvSoftplus(y)

y
x_recovered
```

---

summary.varmxid

*Summary Method for Object of Class varmxid*


---

### Description

Summary Method for Object of Class varmxid

### Usage

```
## S3 method for class 'varmxid'
summary(
  object,
  means = FALSE,
  mu = TRUE,
```

```

    alpha = TRUE,
    beta = TRUE,
    nu = TRUE,
    psi = TRUE,
    theta = TRUE,
    digits = 4,
    ncores = NULL,
    ...
)

```

### Arguments

|                     |                                                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code> | an object of class <code>varmxid</code> .                                                                                                                           |
| <code>means</code>  | Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.                                                                |
| <code>mu</code>     | Logical. If <code>mu = TRUE</code> , include estimates of the mu vector, if available. If <code>mu = FALSE</code> , exclude estimates of the mu vector.             |
| <code>alpha</code>  | Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector. |
| <code>beta</code>   | Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.     |
| <code>nu</code>     | Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.             |
| <code>psi</code>    | Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.         |
| <code>theta</code>  | Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix. |
| <code>digits</code> | Integer indicating the number of decimal places to display.                                                                                                         |
| <code>ncores</code> | Positive integer. Number of cores to use.                                                                                                                           |
| <code>...</code>    | further arguments.                                                                                                                                                  |

### Value

Returns means or raw estimates depending the the value of the argument `means`.

### Author(s)

Ivan Jacob Agaloos Pesigan



**Description**

Sampling Covariance Matrix of the Parameter Estimates

**Usage**

```
## S3 method for class 'varmxid'
vcov(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  robust = FALSE,
  ncores = NULL,
  ...
)
```

**Arguments**

|        |                                                                                                                                               |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class varmxid.                                                                                                                      |
| mu     | Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.                   |
| alpha  | Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector.       |
| beta   | Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.           |
| nu     | Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.                   |
| psi    | Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.               |
| theta  | Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.       |
| robust | Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix. |
| ncores | Positive integer. Number of cores to use.                                                                                                     |
| ...    | additional arguments.                                                                                                                         |

**Value**

Returns a list of sampling variance-covariance matrices.

**Author(s)**

Ivan Jacob Agaloos Pesigan

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